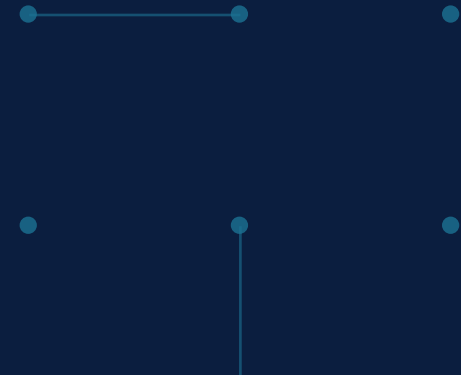




OPTIMAL STRATIFICATION OF A SAMPLING FRAME

# Classical, Quantum, and Quantum-Inspired Approaches Compared

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A comparative case study of a genetic algorithm, D-Wave (simulated annealing), IBM Quantum (QAOA), QCi Dirac-3, and Fixstars Amplify AE on a common twenty-stratum frame from the swissmunicipalities dataset.

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# Why stratification is a hard combinatorial problem

Optimal stratification aggregates many atomic strata into a small number of final strata so as to minimise the total sample size required to meet precision (CV) targets.

1

## Combinatorial

The search space of assignments of  $L$  atomic strata to  $K$  clusters grows like  $K^L$  — exhaustive search is infeasible beyond small  $L$ .

2

## Not simply pairwise

The true objective — the Bethel sample size — is defined implicitly through an allocation problem that couples all strata in the same cluster.

3

## Not a low-degree polynomial

This is exactly why the Bethel objective cannot be encoded directly as a QUBO — every quantum route must substitute a surrogate.

# The Bethel objective must be approximated to reach a QUBO

## TRUE OBJECTIVE

### Bethel–Chromy sample size

Solves a convex allocation problem per candidate partition, coupling all strata in a cluster.

Evaluated directly, at every step, by the classical genetic algorithm (SamplingStrata).

Not a low-degree polynomial in the assignment variables → no direct QUBO form.



## SURROGATE ACTUALLY SOLVED

### Population-weighted pairwise dispersion QUBO

One-hot binary indicators  $x_{hk}$  for “stratum  $h$  in cluster  $k$ ”, with a quadratic penalty enforcing exactly one cluster per stratum.

Minimises within-cluster distance, not the Bethel sample size — a proxy, not the target.

Solved by all four solvers; the Bethel allocation is always recomputed classically afterwards.

# The classical benchmark, and four surrogate-based solvers

|                             |                                           |                                                                                                                      |
|-----------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>SamplingStrata GA</b>    | CLASSICAL BENCHMARK                       | Genetic algorithm evaluating the true Bethel sample size at every step. The practical reference for the whole study. |
| <b>D-Wave (formulation)</b> | SOLVED WITH CLASSICAL SIMULATED ANNEALING | QPU access unavailable on the free plan; same one-hot QUBO solved classically.                                       |
| <b>IBM Quantum</b>          | GATE-BASED · QAOA                         | Binary-label Hamiltonian on ibm_kingston (156 qubits); routing on a sparse heavy-hex lattice is the key constraint.  |
| <b>QCi Dirac-3</b>          | PHOTONIC · ENTROPY QUANTUM COMPUTING      | All-to-all connectivity, one-hot QUBO submitted directly; no embedding, no SWAP overhead.                            |
| <b>Fixstars Amplify AE</b>  | GPU ISING MACHINE · CLASSICAL CONTROL     | Not a quantum device at all — minimises the same QUBO on GPU hardware, isolating the cost of the surrogate.          |

## CASE STUDY

# One common frame: twenty atomic strata, every platform's free tier

20

non-empty atomic strata  
(from a 5×5 cross-classification)

$K = 4$

target number of  
final strata

5%

coefficient-of-variation  
constraint, per target variable

swissmunicipalities dataset (region REG=1, 589 municipalities), discretised by total population (POPTOT) and total area (HApoly) into 5 classes each. Target variables: built-up area (Airbat) and wooded area (Surfacesbois).

## Why exactly twenty strata?

**IBM:** circuit depth on a sparse heavy-hex lattice, not qubit count, is binding.

**Dirac-3:** free-tier cap of 100 decision variables at degree 2 (one-hot needs  $L \times K$ ).

**D-Wave:** no hard ceiling, but run on the same frame for comparability.

### BINDING CONSTRAINT PER SOLVER

#### Amplify AE

Free plan timeout (~5s run)

#### D-Wave (SA)

QPU not authorised on free plan

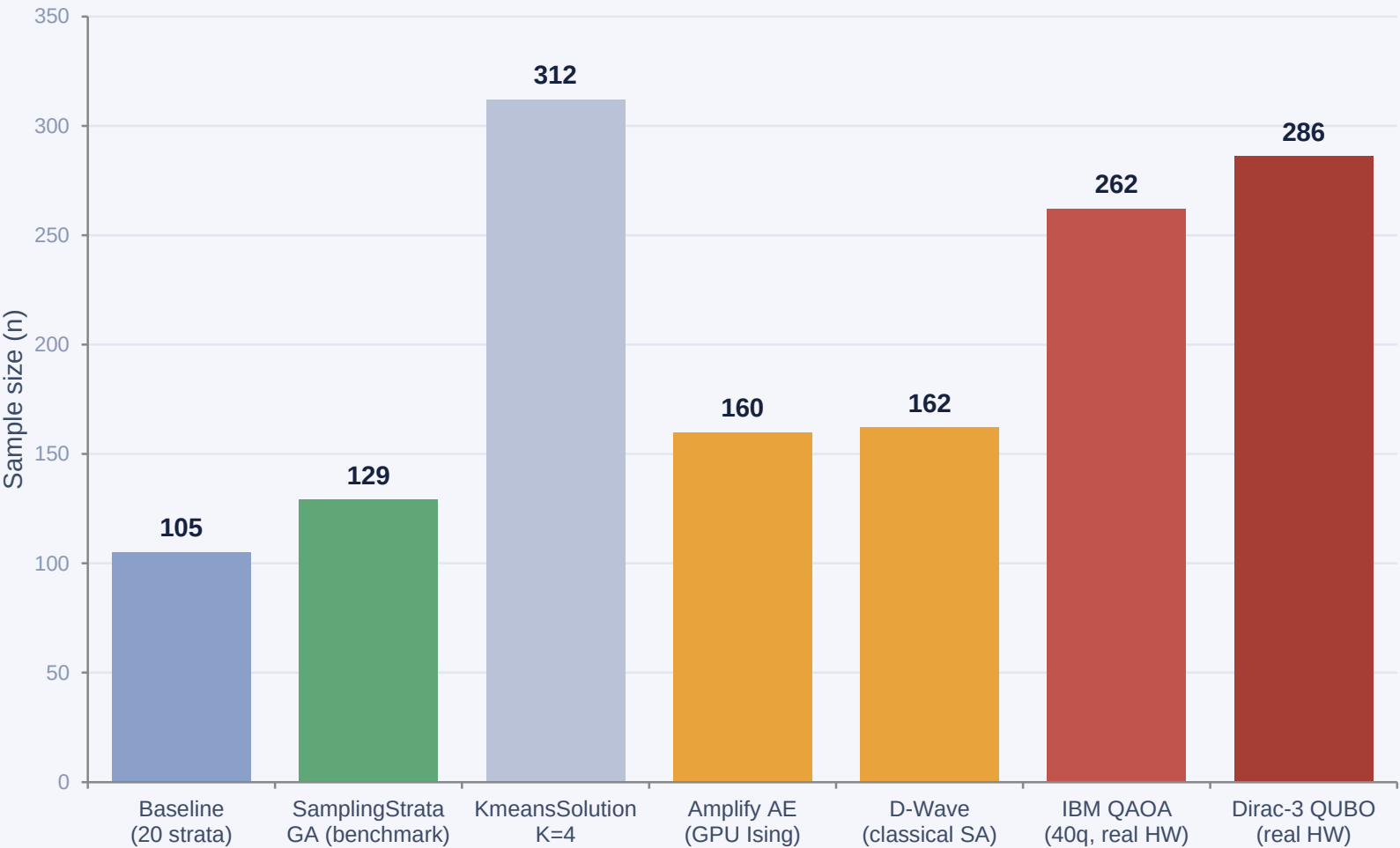
#### IBM Quantum

10 min QPU / 28-day window; no sessions

#### QCi Dirac-3

100-variable cap at polynomial degree 2

# Sample size by method — lower is better



READING THE CHART

**129** GA benchmark — evaluates the true objective directly.

**160** best-known QUBO solution (Amplify AE) — +24% vs. GA: the cost of the surrogate.

**262 / 286** real quantum hardware (IBM / Dirac-3) — fall well short of even the best-known QUBO value.

# Two distinct modes of poor performance

## IBM QUANTUM — QAOA

### Failed to optimise

- Only 2 COBYLA iterations (free-tier time budget)
- Circuit depth ~4,900, ~5,300 two-qubit gates after SWAP routing on the sparse heavy-hex lattice
- Cost energy sat at zero throughout — exactly what the Hamiltonian predicts for a near-uniform, unoptimised output
- Result (262) sits near the 12th percentile of 10,000 random four-cluster assignments — barely better than random search

## QCI DIRAC-3 — PHOTONIC

### Optimised, but settled far short

- Genuine non-trivial optimisation: negative native energy, 10 distinct low-energy solutions returned
- All-to-all photonic connectivity — no embedding, no SWAP overhead, no variational loop
- Re-evaluated under the common QUBO objective: energy 936.61 vs. best-known 501.0 — nearly double
- Searched the right landscape, but the solver settled on a substantially weaker feasible solution

# Two components explain the whole ordering



Both components are empirical, not certified: they rest on the best solutions actually found, not on proofs of optimality.



WOULD PAID ACCESS CHANGE THE PICTURE?

# Paid access relieves some constraints — not the core one

PARTIAL

## D-Wave

Buys real QPU access, but the annealer's sparse connectivity (15–20 neighbours/qubit) likely still limits scaling; hybrid solvers decompose classically, shrinking the quantum share.

TIME, NOT DEPTH

## IBM Quantum

Removes the iteration cap, enabling a fair QAOA test — but the circuit-depth problem is physical: ~5,000 two-qubit gates still accumulate error on superconducting hardware regardless of budget.

LIFTS THE LIMIT

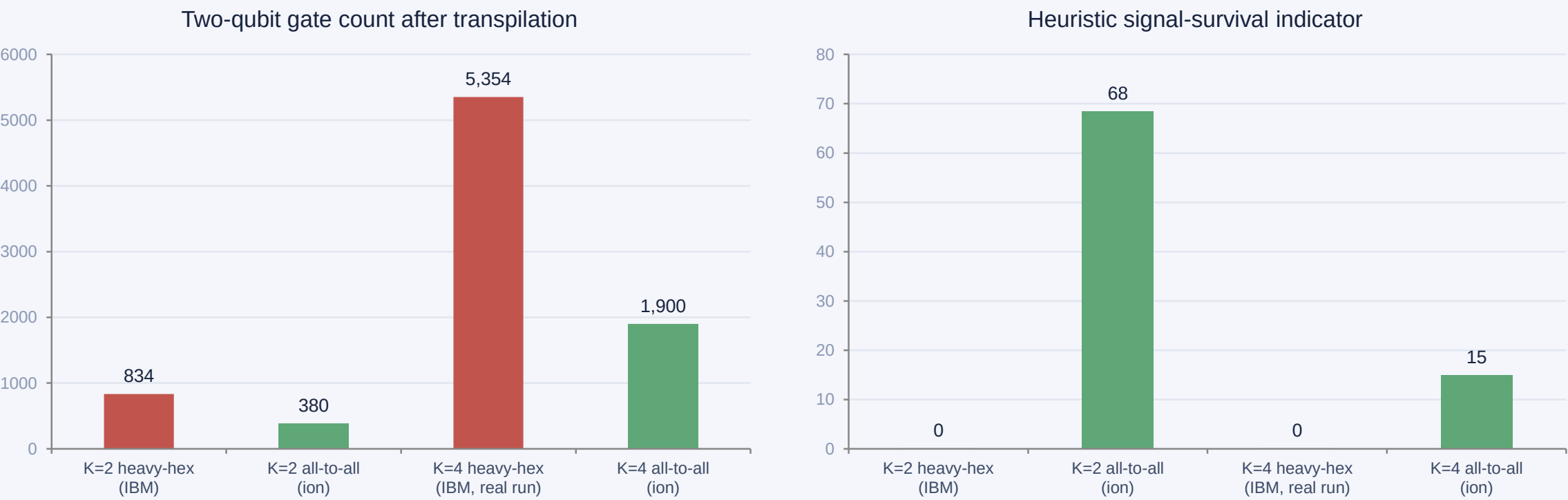
## QCi Dirac-3

The 100-variable cap here was an explicit free-tier quota. Full device capacity (~954 qudit levels) would reach roughly 119 atomic strata at K=4 — the biggest gain of the three.

*What paid access would not change: the formulation gap. Every solver still optimises the surrogate, not the Bethel objective — the best QUBO result stays around 160 vs. 129.*

# Isolating the SWAP-routing hypothesis on trapped ions

IBM's failure is consistent with SWAP-routing overhead on a sparse heavy-hex lattice. Trapped-ion processors (IonQ, Quantinuum) offer effectively all-to-all connectivity with no SWAP overhead. This was tested offline, for free, by transpiling the identical QAOA circuit against a heavy-hex map and a fully-connected map.



A simplified proxy ( $\text{fidelity}^{\text{gate-count}}$ ), not a computed success probability — but it matches the flat, energy-zero outcome actually observed on the real IBM run.

## CONCLUSIONS

# At this scale, classical remains the method of choice

- 1 SamplingStrata's genetic algorithm sets the benchmark at  $n = 129$ , evaluating the true Bethel objective directly.
- 2 The best QUBO solution found by any solver — including a free classical GPU control — maps to  $n = 160$ : a ~24% formulation gap that no solver in this study closed.
- 3 Real quantum hardware (IBM QAOA, QCi Dirac-3) falls further short still (262, 286), for reasons now reasonably well characterised: truncated optimisation and SWAP-routing depth for IBM; a genuine but under-performing search for Dirac-3.
- 4 Two obstacles, in order of priority: (1) bring the encoded objective closer to the true Bethel sample size; (2) only then does hardware quality become the binding question.